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(Vapour + liquid) equilibrium of (DIPE + IPA + water) at 101.32 kPa

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Abstract

Thermodynamically consistent (vapour + liquid) equilibrium data at 101.32 kPa have been determined for (diisopropyl ether + isopropyl alcohol + water) and its constituents (diisopropyl ether + isopropyl alcohol) and (isopropyl alcohol + water). The NRTL and UNIQUAC equations for the liquid phase activity coefficients were found to correlate better the experimental data. The ASOG and the original and modified UNIFAC group-contribution methods did not represent adequately the (vapour + liquid) equilibrium data of this study.

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1. Introduction

Tertiary ethers with 5 or 6 carbons, like diisopropyl ether (DIPE), show excellent antiknock properties besides their less polluting effects, and are becoming the preferred oxygenates for use in gasoline. DIPE is obtained by a reaction of propylene with isopropyl alcohol (IPA), which is initially produced by hydration of propylene. The ether purification involves extraction of alcohol with water. Phase equilibrium data of oxygenated mixtures are important for predicting the vapour phase composition in equilibrium with hydrocarbon mixtures. The present work reports on

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isobaric (vapour + liquid) equilibrium (v.l.e.) data for homogeneous mixtures of (DIPE + IPA + water) and the constituent binary systems at 101.32 kPa. The thermodynamic consistency of the binary data was checked using the tests of Fredenslund [1] and Wisniak-LW [2] and that of the ternary data using both Wisniak and McDermott-Ellis (modified by Wisniak and Tamir) [3,4]. The results were correlated using the Wilson [5], NRTL [6], and UNIQUAC [7] equations and compared with the predictions of ASOG [8,9], UNIFAC [1], UNIFAC-Dortmund [10,11], and UNIFAC-Lyngby [12] group contribution methods.

2. Experimental

Water was purified using a Milli-Q plus system. DIPE and IPA were supplied by Aldrich and have mass fraction purities >0.99 and >0.995 , respectively. The purities were verified chromatographically. The mass fraction water content of DIPE and IPA, determined with a Metrohm737 KF coulometer, were $2 \cdot 10^{-4}$ and $4 \cdot 10^{-4}$, respectively. Table 1 shows the experimental densities and refractive indices at $T = 298.15$ K and boiling temperatures at 101.32 kPa, together with the corresponding literature values [13].

The v.l.e. data were determined in a Labodest 602 distillation apparatus that recycles both liquid and vapour phases (Fischer Labor und Verfahrenstechnik, Germany). This still was equipped with a Fischer digital manometer and Heareus Quat100 quartz thermometer that measured with accuracies of ± 0.01 kPa and ± 0.02 K, respectively. The distillation was carried out at 101.32 kPa under inert argon atmosphere. The liquid and vapour phase compositions were determined indirectly by densimetry and refractometry, together with the published [14] data for the composition dependence of the densities and refractive indices of the systems studied. Densities were measured to within $\pm 0.00001 \text{ g} \cdot \text{cm}^{-3}$ by an Anton Paar DMA 60/602 densimeter. The refractive indices were measured to an accuracy of $\pm 4 \cdot 10^{-5}$ by an ATAGO RX-5000 refractometer. A Hetoterm thermostat was used to maintain the temperature at (298.15 ± 0.02) K. For some compositions of the ternary system the density and refractive index isolines were far from orthogonal, and in fact, overlapped over rather wide composition intervals. The indirect method was not sufficiently precise and mixtures in these intervals were analysed directly by capillary gas chromatography, using a Hewlett-Packard Series II chromatograph (model

TABLE 1
Density (ρ) and refractive index (n_D) at $T = 298.15$ K and boiling temperature (T) of the pure compounds

Compound	$\rho/(\text{g} \cdot \text{cm}^{-3})$		n_D		T/K	
	Expt	Lit [13]	Expt	Lit [13]	Expt	Lit [13]
Water	0.99704	0.99704	1.33250	1.33250	373.15	373.15
IPA	0.78095	0.78126	1.37501	1.3752	355.32	355.392
DIPE	0.71845	0.71854	1.36512	1.3655	341.36	341.66

5890) equipped with an HP-FFAP capillary column and a thermal conductivity detector linked to an HP Series II integrator (model 3396). The maximum mole fraction deviation in composition with both the direct and indirect methods was ± 0.003 .

3. Results

The experimental liquid and vapour phase compositions (x_i and y_i , respectively) and equilibrium temperatures (T), together with the corresponding activity coefficients (γ_i) and excess Gibbs energies (G^E/RT) for the binary systems are listed in table 2 and the same parameters for the ternary system are listed in table 3. Measurements of the ternary system are limited to the miscible region, the boundaries of which were established in a previous work [14]. The v.l.e. data for (DIPE + IPA) and (IPA + water) are compared with data reported by other authors [14–18] in figure 1. Figure 2 shows the isotherms of the liquid phase for v.l.e. of (DIPE + IPA + water). Yorzane *et al.* [17] and Verhoeve [18] have also published data on the ternary system which are in agreement with our measurements.

4. Data treatment

For the vapour and liquid phases in equilibrium at temperature T , and pressure p

$$y_i \phi_i p = x_i \gamma_i p_i^s \phi_i^s \exp \left\{ \frac{V_i^L (p - p_i^s)}{RT} \right\}, \quad (1)$$

where p_i^s is the saturated vapour pressure of component i as calculated from Antoine equation

$$\log(p_i^s/\text{kPa}) = A - \frac{B}{(T/\text{K}) + C}, \quad (2)$$

with the coefficients A , B and C taken from the literature [17,19] and are shown in table 4, V_i^L is the molar volume of component i in the liquid phase calculated using the Rackett equation, and ϕ_i and ϕ_i^s are the fugacity coefficient and fugacity coefficient at saturation of component i , respectively, and were calculated from the second virial coefficient by the method of Hayden and O'Connell [20].

The thermodynamic consistency of the v.l.e. data for the binary systems was checked by using both Fredeslund's test (mean deviation between calculated and experimental $y_i < 0.01$ for both systems) and Wisniak's L-W test ($D < 2$). The consistency of the ternary data was also checked using two tests: Wisniak's L-W test ($0.94 < L_i/W_i < 1.00$ for all data points) and Wisniak-Tamir's modification of the McDermott-Ellis test ($D < M_{\max}$ at all data points).

4.1. Correlation

The correlation of the experimental (p, T, x, y) results was performed by using a non-linear regression method based on the maximum likelihood principle. The

TABLE 2

Boiling temperatures (T), liquid and vapour mole fractions (x_i , y_i), and activity coefficients (γ_i), G^E/RT for the binary systems

T/K	x_1	y_1	γ_1	γ_2	G^E/RT	T/K	x_1	y_1	γ_1	γ_2	G^E/RT
DIPE(1) + IPA(2)											
354.12	0.0135	0.0564	2.9927	1.0032	0.0180	339.84	0.6135	0.7070	1.2214	1.4798	0.2742
352.38	0.0396	0.1578	2.9818	0.9860	0.0298	339.63	0.6613	0.7263	1.1711	1.5942	0.2624
349.72	0.0894	0.2961	2.6560	0.9700	0.0596	339.49	0.6920	0.7385	1.1426	1.6868	0.2533
346.92	0.1612	0.4100	2.2013	0.9944	0.1225	339.41	0.7257	0.7526	1.1127	1.7998	0.2387
346.39	0.1801	0.4365	2.1283	0.9945	0.1315	339.37	0.7486	0.7636	1.0956	1.8810	0.2272
344.91	0.2296	0.4955	1.9753	1.0114	0.1651	339.35	0.7904	0.7849	1.0669	2.0572	0.2024
343.79	0.2773	0.5362	1.8273	1.0422	0.1970	339.37	0.8133	0.7980	1.0533	2.1684	0.1868
343.18	0.3075	0.5574	1.7433	1.0669	0.2157	339.44	0.8356	0.8127	1.0416	2.2779	0.1694
342.71	0.3349	0.5759	1.6763	1.0875	0.2288	339.51	0.8642	0.8341	1.0312	2.4377	0.1476
342.25	0.3636	0.5940	1.6138	1.1113	0.2412	339.71	0.8932	0.8594	1.0214	2.6068	0.1213
341.80	0.3935	0.6098	1.5511	1.1442	0.2544	340.00	0.9266	0.8928	1.0136	2.8596	0.0896
341.88	0.3917	0.6083	1.5507	1.1410	0.2521	340.32	0.9484	0.9184	1.0086	3.0561	0.0658
340.72	0.4932	0.6594	1.3810	1.2575	0.2753	340.58	0.9651	0.9416	1.0081	3.2004	0.0484
340.37	0.5320	0.6746	1.3233	1.3227	0.2799	340.97	0.9847	0.9708	1.0066	3.5927	0.0261
340.08	0.5741	0.6924	1.2693	1.3934	0.2782						
IPA(1) + water(2)											
354.94	0.9777	0.9638	1.0012	3.2192	0.0272	353.75	0.4868	0.6020	1.3177	1.6044	0.3769
354.66	0.9545	0.9316	1.0022	3.0127	0.0522	354.02	0.4231	0.5853	1.4586	1.4711	0.3824
354.40	0.9313	0.9011	1.0037	2.9135	0.0769	354.22	0.3791	0.5791	1.5981	1.3762	0.3760
354.14	0.9037	0.8678	1.0064	2.8056	0.1051	354.44	0.3413	0.5716	1.7371	1.3089	0.3658
353.80	0.8623	0.821	1.0114	2.6910	0.1461	354.73	0.2874	0.5634	2.0105	1.2189	0.3418
353.64	0.8319	0.7929	1.0190	2.5657	0.1741	355.00	0.2376	0.5551	2.3710	1.1486	0.3107
353.49	0.8000	0.7652	1.0288	2.4586	0.2026	356.18	0.1114	0.5175	4.5051	1.0201	0.1853
353.38	0.7631	0.7365	1.0428	2.3387	0.2332	358.70	0.0545	0.4565	7.3840	0.9790	0.0889
353.29	0.7089	0.6994	1.0699	2.1780	0.2745	364.37	0.0251	0.3152	9.0093	0.9650	0.0204
353.28	0.6895	0.6877	1.0821	2.1220	0.2880	366.79	0.0172	0.2345	8.9912	0.9785	0.0164
353.28	0.6713	0.6772	1.0946	2.0716	0.3001	369.04	0.0108	0.1555	8.7966	0.9883	0.0118
353.32	0.6409	0.6619	1.1189	1.9826	0.3178	370.64	0.0068	0.0976	8.3139	0.9931	0.0075
353.39	0.6047	0.6412	1.1458	1.9056	0.3372	371.95	0.0034	0.0502	8.1924	0.9943	0.0014
353.52	0.5513	0.6227	1.2144	1.7559	0.3597						

TABLE 3

Boiling temperatures (T), liquid and vapour mole fractions (x_i , y_i), activity coefficients (γ_i) and excess molar Gibbs energies (G^E/RT) for {DIPE(1) + IPA(2) + water(3)} at 101.32 kPa

x_1	x_2	y_1	y_2	T/K	γ_1	γ_2	γ_3	G^E/RT
0.7022	0.2427	0.7035	0.1768	336.86	1.1651	1.6312	9.4093	0.3496
0.6575	0.2764	0.6852	0.1874	337.03	1.2061	1.5047	8.2734	0.3758
0.7078	0.2432	0.7005	0.1784	337.07	1.1435	1.6262	10.6017	0.3289
0.6737	0.2722	0.6815	0.1888	337.17	1.1658	1.5289	10.2238	0.3447
0.7244	0.2343	0.7122	0.1795	337.19	1.1314	1.6899	11.2007	0.3121
0.6824	0.2645	0.6939	0.1873	337.19	1.1707	1.5605	9.5425	0.3450
0.7094	0.2475	0.7059	0.1842	337.25	1.1431	1.6365	10.8575	0.3196
0.6429	0.2911	0.6786	0.2084	337.62	1.1995	1.5450	7.1579	0.3735
0.7641	0.2182	0.7470	0.1899	337.97	1.0970	1.8549	14.7577	0.2532
0.5724	0.3567	0.6613	0.2535	338.38	1.2826	1.4795	4.8554	0.3942
0.5136	0.3678	0.6470	0.2176	338.41	1.3982	1.2287	4.5970	0.4288
0.8546	0.1394	0.8038	0.1552	338.43	1.0394	2.3304	27.8399	0.1709
0.3687	0.4885	0.5925	0.2491	338.82	1.7643	1.0366	4.3706	0.4375
0.5160	0.3839	0.6363	0.2699	338.84	1.3506	1.4311	3.7036	0.4238
0.4617	0.4202	0.6209	0.2595	338.86	1.4729	1.2548	3.9930	0.4377
0.4684	0.4226	0.6202	0.2687	338.90	1.4484	1.2896	4.0125	0.4324
0.4759	0.4229	0.6224	0.2758	338.90	1.4304	1.3229	3.9614	0.4280
0.3343	0.4940	0.5774	0.2520	338.98	1.8880	1.0287	3.8832	0.4594
0.4057	0.4718	0.6022	0.2601	339.25	1.6075	1.0993	4.3503	0.4173
0.5188	0.3989	0.6314	0.3080	339.25	1.3162	1.5423	2.8597	0.4018
0.4583	0.4347	0.6056	0.3059	339.27	1.4295	1.4025	3.2025	0.4353
0.4890	0.4313	0.6167	0.3180	339.27	1.3636	1.4704	3.1762	0.4101
0.4001	0.4926	0.5996	0.2764	339.57	1.6070	1.1025	4.4108	0.3971
0.4879	0.4363	0.6252	0.3127	339.62	1.3704	1.4071	3.1292	0.3892
0.3178	0.5083	0.5644	0.2730	339.74	1.8972	1.0455	3.5326	0.4456
0.2746	0.5417	0.5435	0.2728	340.39	2.0745	0.9509	3.6668	0.4118
0.3213	0.5409	0.5624	0.3028	340.41	1.8316	1.0571	3.5911	0.4006
0.4102	0.5400	0.6170	0.3055	340.58	1.5626	1.0627	5.6947	0.3026
0.2382	0.5482	0.5244	0.2873	340.68	2.2888	0.9760	3.1887	0.4316
0.2138	0.4940	0.5115	0.2823	340.86	2.4752	1.0551	2.5302	0.4915
0.2976	0.5727	0.5444	0.3487	341.11	1.8745	1.1133	2.9342	0.3881
0.1943	0.4822	0.4917	0.2992	341.33	2.5830	1.1210	2.2687	0.5045
0.3336	0.5810	0.5633	0.3461	341.36	1.7159	1.0779	3.7410	0.3364
0.3783	0.5529	0.5893	0.3374	341.43	1.5783	1.1020	3.7520	0.3173
0.1701	0.5204	0.4857	0.2953	341.72	2.8812	1.0072	2.4410	0.4599
0.3084	0.5916	0.5480	0.3519	341.80	1.7829	1.0547	3.4598	0.3339
0.2654	0.5649	0.5240	0.3078	341.86	1.9801	0.9624	3.4079	0.3677
0.2635	0.5756	0.5240	0.3078	342.13	1.9782	0.9332	3.5527	0.3440
0.1574	0.5815	0.4761	0.3133	342.15	3.0135	0.9379	2.7307	0.3987
0.2654	0.5895	0.5302	0.3170	342.21	1.9818	0.9354	3.5688	0.3268
0.2692	0.5961	0.5256	0.3413	342.29	1.9322	0.9923	3.3379	0.3351
0.2805	0.6122	0.5251	0.3709	342.29	1.8522	1.0501	3.2762	0.3302
0.1346	0.5295	0.4629	0.3093	342.37	3.4059	1.0066	2.2724	0.4441

TABLE 3 (continued)

x_1	x_2	y_1	y_2	T/K	γ_1	γ_2	γ_3	G^E/RT
0.2731	0.6010	0.5347	0.3239	342.43	1.9290	0.9285	3.7721	0.3020
0.2330	0.5897	0.4988	0.3307	342.45	2.1110	0.9641	3.2197	0.3598
0.3240	0.6073	0.5590	0.3659	342.66	1.6860	1.0287	3.6450	0.2753
0.1953	0.7814	0.4785	0.4839	343.39	2.3480	1.0213	5.2004	0.2216
0.1841	0.6406	0.4422	0.3987	343.63	2.2908	1.0141	2.8831	0.3472
0.1211	0.5297	0.4205	0.3237	343.92	3.2894	0.9825	2.2924	0.4245
0.2013	0.7223	0.4594	0.4737	344.34	2.1280	1.0368	2.7061	0.2542
0.0574	0.3994	0.3986	0.3305	344.34	6.5036	1.3056	1.5315	0.4455
0.1857	0.7472	0.4565	0.4703	344.38	2.2898	0.9932	3.3648	0.2302
0.1032	0.5113	0.3995	0.3377	344.40	3.6185	1.0394	2.0885	0.4364
0.0636	0.4580	0.3942	0.3283	344.63	5.7566	1.1167	1.7591	0.4321
0.0951	0.7032	0.4083	0.3841	344.97	3.9421	0.8388	3.0823	0.2339
0.1785	0.7353	0.4628	0.4237	345.01	2.3707	0.8848	3.9520	0.1826
0.0906	0.6799	0.3796	0.4381	345.07	3.8392	0.9847	2.3676	0.3092
0.0777	0.5172	0.3489	0.3853	345.42	4.0815	1.1206	1.9223	0.4329
0.1441	0.6386	0.3659	0.4537	345.51	2.2979	1.0650	2.4280	0.3529
0.0567	0.4601	0.3568	0.3518	345.52	5.7028	1.1455	1.7591	0.4341
0.0769	0.5198	0.3444	0.3937	345.62	4.0474	1.1295	1.8865	0.4268
0.1637	0.7371	0.4317	0.4546	345.73	2.3635	0.9174	3.3334	0.1967
0.0844	0.7226	0.3598	0.4618	345.74	3.8328	0.9485	2.6770	0.2652
0.0837	0.7424	0.3721	0.4529	345.83	3.9842	0.9021	2.9048	0.2247
0.0780	0.5392	0.3474	0.3570	346.03	3.9780	0.9703	2.2040	0.3940
0.1216	0.8051	0.3913	0.5078	346.09	2.8577	0.9230	3.9387	0.1636
0.0795	0.7236	0.3563	0.4616	346.09	3.9890	0.9326	2.6390	0.2506
0.1068	0.8396	0.3980	0.5049	346.17	3.3007	0.8771	5.1677	0.1055
0.1198	0.6663	0.3417	0.4647	346.33	2.5227	1.0090	2.5552	0.3175
0.0866	0.8617	0.3872	0.5007	346.39	3.9370	0.8394	6.1245	0.0615
0.0777	0.7344	0.3341	0.4884	346.46	3.7892	0.9567	2.6526	0.2543
0.0748	0.8735	0.3763	0.5104	346.48	4.4201	0.8406	6.1644	0.0536
0.0763	0.7556	0.3575	0.4531	346.56	4.1134	0.8593	3.1522	0.1864
0.0628	0.5304	0.3107	0.4159	346.64	4.3464	1.1189	1.8687	0.4062
0.0884	0.8490	0.3452	0.5634	346.76	3.4066	0.9429	4.0569	0.1461
0.1127	0.5795	0.2946	0.4867	346.78	2.2872	1.1910	1.9651	0.4025
0.0729	0.5544	0.3053	0.4086	346.84	3.6592	1.0428	2.1160	0.3971
0.1112	0.5704	0.2928	0.4846	346.88	2.2975	1.1997	1.9253	0.4049
0.0286	0.2698	0.3477	0.3636	347.04	10.5410	1.8920	1.1262	0.3228
0.0674	0.8651	0.3393	0.5430	347.16	4.3429	0.8767	4.7617	0.0905
0.0944	0.5905	0.2766	0.4943	347.20	2.5349	1.1659	1.9748	0.3929
0.0687	0.7691	0.3333	0.4759	347.22	4.1823	0.8619	3.1994	0.1726
0.0880	0.5892	0.2764	0.4766	347.47	2.6967	1.1139	2.0545	0.3833
0.0645	0.5745	0.2726	0.4426	347.63	3.6141	1.0538	2.1028	0.3813
0.1052	0.6525	0.2727	0.5338	347.67	2.2116	1.1171	2.1283	0.3387
0.0491	0.5406	0.2635	0.4531	347.79	4.5698	1.1387	1.8286	0.3924
0.0712	0.6024	0.2410	0.5038	347.99	2.8676	1.1263	2.0527	0.3814
0.0660	0.8409	0.2791	0.6123	348.23	3.5455	0.9714	3.0427	0.1627

TABLE 3 (continued)

x_1	x_2	y_1	y_2	T/K	γ_1	γ_2	γ_3	G^E/RT
0.0585	0.5900	0.2290	0.4968	348.35	3.2847	1.1170	2.0167	0.3814
0.0235	0.2834	0.2973	0.3748	348.45	10.5583	1.7488	1.2190	0.3511
0.1053	0.6175	0.2283	0.5692	348.51	1.8093	1.2145	1.8783	0.3572
0.0759	0.8064	0.2612	0.6117	348.57	2.8600	0.9974	2.7754	0.1978
0.0291	0.4580	0.2093	0.4695	349.12	5.9121	1.3170	1.5669	0.4082
0.0556	0.6104	0.1747	0.5541	349.44	2.5618	1.1505	2.0053	0.3703
0.0564	0.8259	0.2372	0.5989	349.58	3.4002	0.9142	3.4297	0.1400
0.0277	0.5054	0.1856	0.4763	349.78	5.4120	1.1781	1.7625	0.3942
0.0202	0.2950	0.2275	0.4096	349.80	9.0757	1.7351	1.2894	0.3812
0.0264	0.4712	0.1835	0.4732	349.84	5.6056	1.2523	1.6589	0.4058
0.0193	0.4325	0.1685	0.4711	350.29	6.9580	1.3335	1.5663	0.4079
0.0261	0.5191	0.1596	0.4890	350.43	4.8556	1.1466	1.8304	0.3872
0.0228	0.4865	0.1383	0.5260	350.57	4.8016	1.3083	1.6113	0.4006
0.0125	0.3920	0.1656	0.4681	350.60	10.4677	1.4436	1.4470	0.3933
0.0198	0.2857	0.1618	0.4552	350.81	6.4211	1.9098	1.2860	0.3964
0.0240	0.5348	0.1261	0.5322	350.95	4.1173	1.1857	1.7958	0.3833
0.0171	0.4626	0.1232	0.5174	351.21	5.6066	1.3187	1.5845	0.3969
0.0378	0.9228	0.1926	0.7193	351.27	3.9308	0.9167	5.1436	0.0360
0.0218	0.5503	0.1000	0.5628	351.38	3.5557	1.1974	1.7952	0.3772
0.0145	0.3956	0.1276	0.4945	351.47	6.7980	1.4586	1.4539	0.3979
0.0108	0.4102	0.0892	0.5425	352.00	6.2974	1.5104	1.4127	0.3891
0.0083	0.3450	0.0751	0.5440	352.63	6.7839	1.7561	1.2751	0.3673
0.0090	0.4350	0.0602	0.5421	352.80	4.9966	1.3786	1.5376	0.3933
0.0109	0.4213	0.0677	0.5471	352.85	4.6302	1.4336	1.4557	0.3817
0.0063	0.3583	0.0606	0.5436	353.02	7.1407	1.6637	1.3273	0.3747
0.0050	0.3701	0.0482	0.5435	353.34	7.0987	1.5902	1.3743	0.3802
0.0058	0.3882	0.0359	0.5521	353.58	4.5308	1.5256	1.4162	0.3836

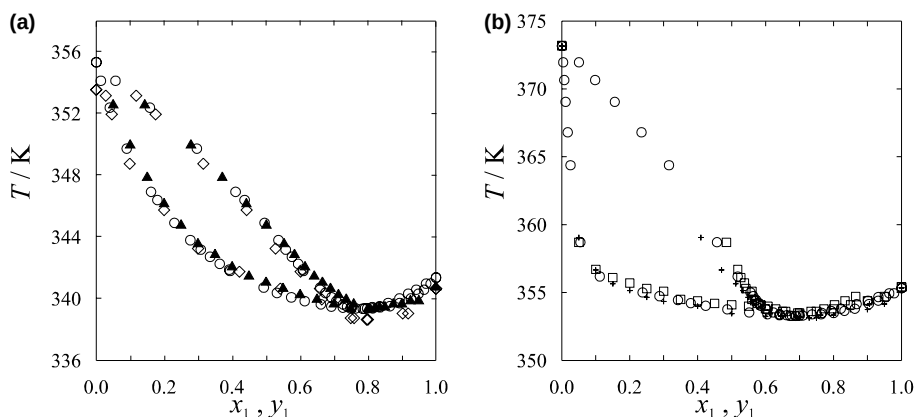


FIGURE 1. Temperature-composition (mole fraction) diagram for (DIPE + IPA) (a) and (IPA + water) (b): (○), this work; (▲), Yorizane *et al.* [17]; (◇), Verhoye [18]; (+), Rajendran *et al.* [15], (□), Udovenko *et al.* [16].

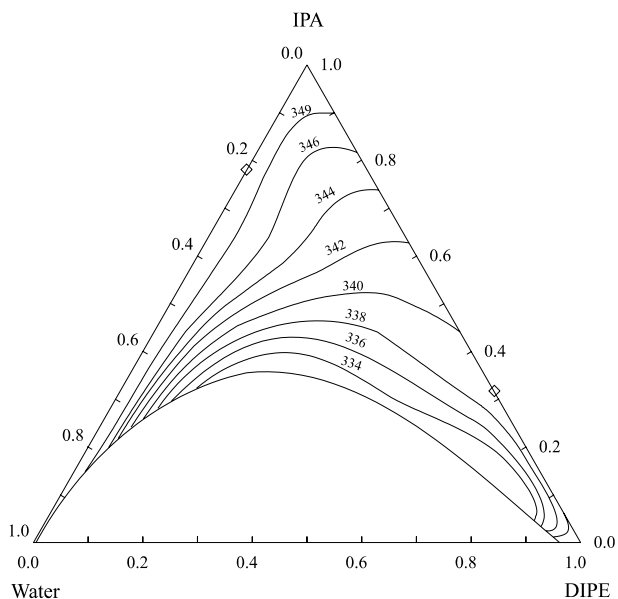


FIGURE 2. Temperature isolines (K) for (DIPE + IPA + water) at 101.32 kPa; ◊, indicate binary azeotropes.

TABLE 4
Antoine coefficients A , B , C for equation (2)

Compound	A	B	C	Reference
DIPE	5.97678	1257.6	−43.15	Reid <i>et al.</i> [19]
IPA	5.7853	813.055	−140.22	Yorizane <i>et al.</i> [17]
Water	7.07405	1657.459	−46.13	Reid <i>et al.</i> [19]

TABLE 5
Activity model parameters and root mean square deviations (σ): Wilson, NRTL, and UNIQUAC models

Model	$\sigma(T)/K$ $\sigma(x)$ $\sigma(y)$ $\sigma(p)/kPa$					
DIPE(1) + IPA(2)						
Wilson	$\frac{\Delta\lambda_{12}}{J\cdot mol^{-1}} = -629.852$	$\frac{\Delta\lambda_{21}}{J\cdot mol^{-1}} = 4644.37$	0.17	0.0033	0.0042	0.08
NRTL ($\alpha = 0.1$)	$\frac{\Delta g_{12}}{J\cdot mol^{-1}} = 5483.5$	$\frac{\Delta g_{21}}{J\cdot mol^{-1}} = -1712.19$	0.15	0.0036	0.0035	0.07
UNIQUAC	$\frac{\Delta u_{12}}{J\cdot mol^{-1}} = 7241.74$	$\frac{\Delta u_{21}}{J\cdot mol^{-1}} = -1890.6$	0.15	0.0027	0.0030	0.08
IPA(1) + water(2)						
Wilson	$\frac{\Delta\lambda_{12}}{J\cdot mol^{-1}} = 2826.18$	$\frac{\Delta\lambda_{21}}{J\cdot mol^{-1}} = 5301.59$	0.31	0.0048	0.0080	0.17
NRTL ($\alpha = 0.3$)	$\frac{\Delta g_{12}}{J\cdot mol^{-1}} = -0.54682$	$\frac{\Delta g_{21}}{J\cdot mol^{-1}} = 6675.81$	0.20	0.0045	0.0046	0.11
UNIQUAC	$\frac{\Delta u_{12}}{J\cdot mol^{-1}} = 261.949$	$\frac{\Delta u_{21}}{J\cdot mol^{-1}} = 3872.91$	0.22	0.0036	0.0055	0.12

Binary interaction parameters of the Wilson, NRTL and UNIQUAC equations as obtained by correlating the v.l.e. data for {DIPE(1) + IPA(2) + water(3)}, together with the root mean square deviations (σ) in temperature, vapour and liquid equilibrium compositions and pressure

Model			$\sigma(T)/\text{K}$	$\sigma(x_1)$	$\sigma(y_1)$	$\sigma(x_2)$	$\sigma(y_2)$	$\sigma(p)/\text{kPa}$
Wilson	$\frac{\Delta \hat{z}_{12}}{\text{J}\cdot\text{mol}^{-1}} = 207.3096$	$\frac{\Delta \hat{z}_{21}}{\text{J}\cdot\text{mol}^{-1}} = 3813.466$	0.67	0.0169	0.0153	0.0077	0.0201	4.60
	$\frac{\Delta \hat{z}_{13}}{\text{J}\cdot\text{mol}^{-1}} = 24940.34$	$\frac{\Delta \hat{z}_{31}}{\text{J}\cdot\text{mol}^{-1}} = 17678.89$						
	$\frac{\Delta \hat{z}_{23}}{\text{J}\cdot\text{mol}^{-1}} = 2517.645$	$\frac{\Delta \hat{z}_{32}}{\text{J}\cdot\text{mol}^{-1}} = 4782.961$						
NRTL ($\alpha = 0.1$)	$\frac{\Delta g_{12}}{\text{J}\cdot\text{mol}^{-1}} = -865.986$	$\frac{\Delta g_{21}}{\text{J}\cdot\text{mol}^{-1}} = 4622.833$	0.60	0.0161	0.0150	0.0063	0.0189	0.31
	$\frac{\Delta g_{13}}{\text{J}\cdot\text{mol}^{-1}} = 4088.659$	$\frac{\Delta g_{31}}{\text{J}\cdot\text{mol}^{-1}} = 17678.89$						
	$\frac{\Delta g_{23}}{\text{J}\cdot\text{mol}^{-1}} = -5101.64$	$\frac{\Delta g_{32}}{\text{J}\cdot\text{mol}^{-1}} = 12266.48$						
UNIQUAC	$\frac{\Delta u_{12}}{\text{J}\cdot\text{mol}^{-1}} = 6289.541$	$\frac{\Delta u_{21}}{\text{J}\cdot\text{mol}^{-1}} = -1607.84$	0.61	0.0165	0.0150	0.0058	0.0188	0.30
	$\frac{\Delta u_{13}}{\text{J}\cdot\text{mol}^{-1}} = 17755.38$	$\frac{\Delta u_{31}}{\text{J}\cdot\text{mol}^{-1}} = -745.466$						
	$\frac{\Delta u_{23}}{\text{J}\cdot\text{mol}^{-1}} = -99.901$	$\frac{\Delta u_{32}}{\text{J}\cdot\text{mol}^{-1}} = 3838.324$						

models used for the liquid phase activity coefficients were the Wilson equation, NRTL equation, with the nonrandomness parameter α chosen to give the best correlation, and the UNIQUAC equation. For the last equation, the structural parameters r and q

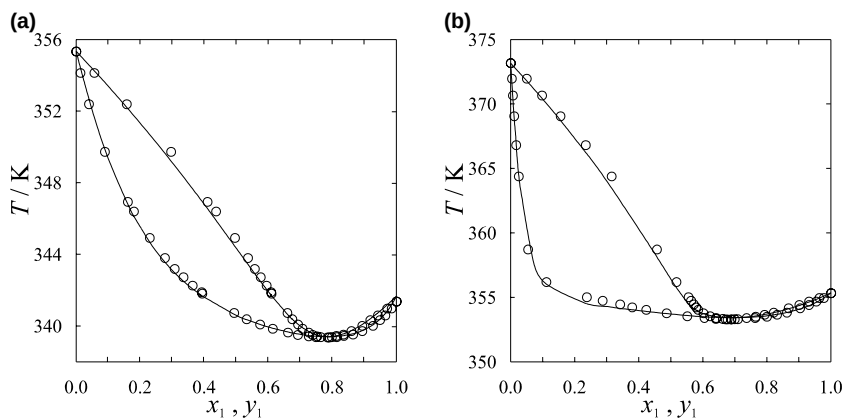


FIGURE 3. Experimental v.l.e. data ($\circ$) and corresponding UNIQUAC and NRTL ($\alpha = 0.3$) results (—) for (DIPE + IPA) (a) and (IPA + water) (b), respectively.

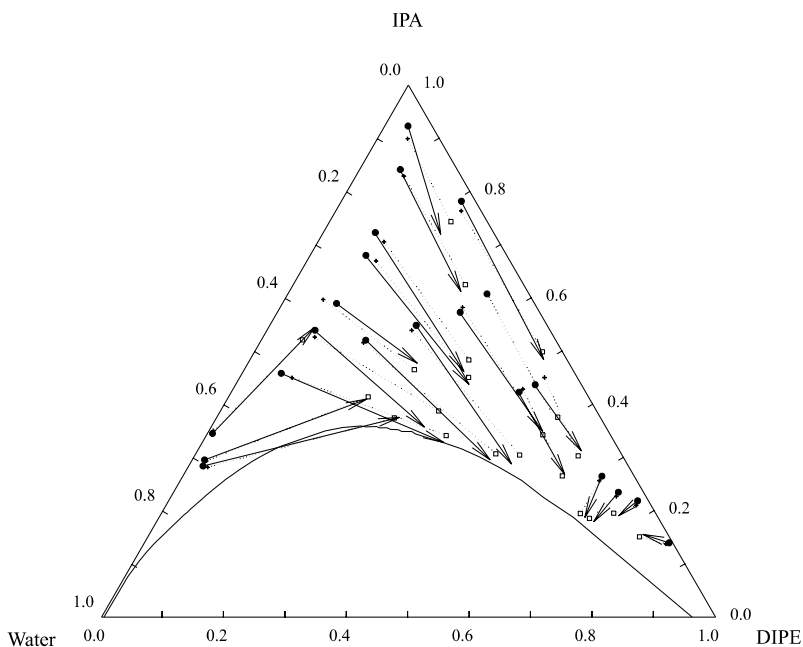


FIGURE 4. Experimental v.l.e. data ($\bullet \rightarrow$) and the corresponding NRTL ($\alpha = 0.1$) results ($+$ — $\square$) for (DIPE + IPA + water).

were taken from Daubert and Danner [21] and the parameter q' was set to 1.00 for water and 0.89 for IPA and was taken from Anderson and Prausnitz [22]. The values of the interaction parameters are summarised in table 5 for the binary systems and in table 6 for the ternary system. Figure 3 compares the UNIQUAC, and the NRTL ($\alpha = 0.3$) temperature-composition curves for (DIPE + IPA) and (IPA + water), respectively. Figure 4 compares the results of (DIPE + IPA + water), the calculated values using the NRTL ($\alpha = 0.1$) equation with the experimental v.l.e. data (for the sake of clarity, the number of data points shown has been reduced).

4.2. Prediction

The v.l.e. data for the binary and ternary systems were calculated using the following group contribution methods for the liquid-phase activity coefficients: the ASOG method, the original UNIFAC method, with the structural and group-interaction parameters recommended by Gmehling *et al.* [23], the UNIFAC-Dortmund method, and the UNIFAC-Lyngby method. Table 7 lists the root mean standard deviations between the experimental v.l.e. data for the binary systems and the calculated values. Figure 5 compares the experimental v.l.e. compositions of

TABLE 7

Root mean square deviations (σ) between the experimental boiling temperatures (T) and vapour phase compositions (y) and those calculated by the ASOG, UNIFAC, and modified UNIFAC methods

System	ASOG		UNIFAC		UNIFAC-Dortmund		UNIFAC-Lyngby	
	$\sigma(T)/K$	$\sigma(y_1)$	$\sigma(T)/K$	$\sigma(y_1)$	$\sigma(T)/K$	$\sigma(y_1)$	$\sigma(T)/K$	$\sigma(y_1)$
DIPE(1) + IPA(2)	0.4	0.015	1.6	0.019	0.1	0.012	0.6	0.008
IPA(1) + water(2)	1.0	0.028	1.1	0.026	0.6	0.016	1.3	0.032

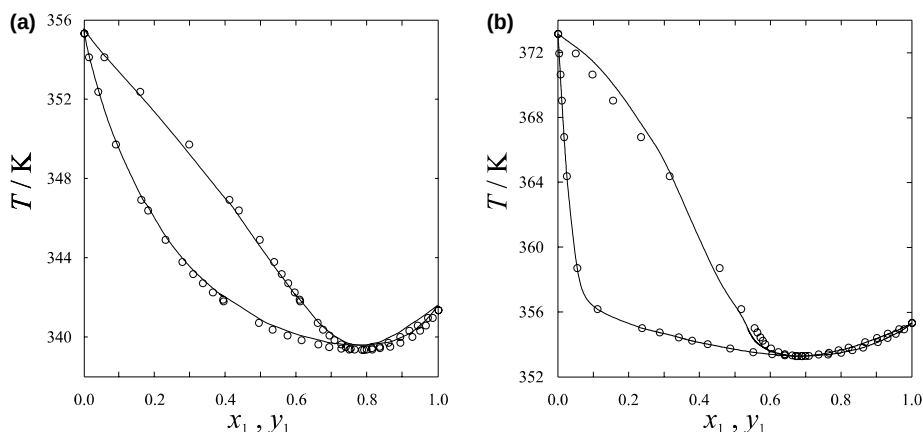


FIGURE 5. Experimental v.l.e. data (○) and corresponding UNIFAC-Dortmund results (—) for (DIPE + IPA) (a) and (IPA + water) (b).

TABLE 8

Root mean square deviations (σ) between the experimental boiling temperatures (T) and vapour phase compositions (y) for {DIPE(1)+IPA(2)+water(3)} and the calculated values by the ASOG, UNIFAC, and modified UNIFAC methods

Model	$\sigma(T_b)/K$	$\sigma(y_1)$	$\sigma(y_2)$	$\sigma(y_3)$
ASOG	2.1	0.057	0.068	0.029
UNIFAC	1.8	0.085	0.082	0.052
UNIFAC-Dortmund	2.1	0.063	0.063	0.024
UNIFAC-Lyngby	2.8	0.067	0.059	0.026

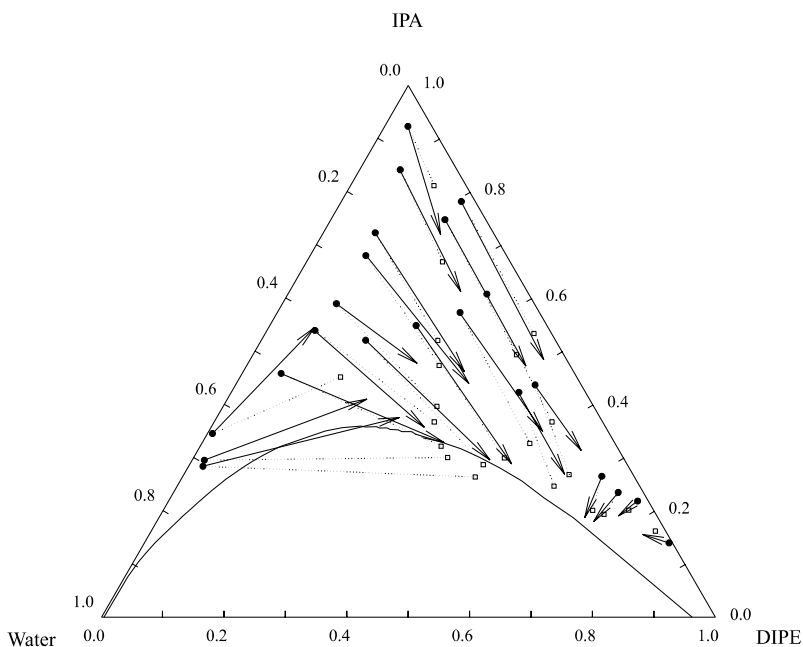


FIGURE 6. Comparison of the experimental v.l.e. data ($\rightarrow$) for (DIPE + IPA + water) with predicted values using the UNIFAC-Dortmund equation ($--\square$).

(DIPE + IPA) and (IPA + water) with the predictions of the UNIFAC-Dortmund model. In table 8 the root mean standard deviations between the experimental v.l.e. data for the ternary system and the calculated ones are presented and in figure 6 the experimental v.l.e. data are compared with the predictions of the UNIFAC-Dortmund model.

5. Conclusions

Thermodynamically consistent isobaric (101.32 kPa) v.l.e. data were determined for (DIPE + IPA + water) and the constituent binary systems. (DIPE + IPA) and

(IPA + water) form azeotropes at the minimum boiling point, and the data are in agreement with those found in the literature. Figures 2 and 4 show the existence of an heteroazeotrope for (DIPE + IPA + water) since the focus of arrows in diagram compositions, point where the isotherm liquid-phase compositions converge, would be found in the immiscible zone of the ternary system.

The Wilson, NRTL and UNIQUAC equations show similar reasonable results in the correlation of the v.l.e. data of the binary systems. However, for the ternary system, The Wilson equation gave a very high deviation in pressure. Therefore, for (DIPE + IPA + water) only NRTL and UNIQUAC yielded reasonable correlations.

Among various group contribution methods used in the prediction of v.l.e. for binary and ternary systems, the UNIFAC-Dortmund method gave the best predictions, however, even with this method the deviations from the experimental data are relatively high.

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